

Exploration strategies for DQN algorithm - report

Wojciech Mierzejek 459435

January 27, 2026

1 Hyperparameter investigation

All test were performed using the original arguments that had been declared in the assignment.

1.1 Epsilon-greedy DQN

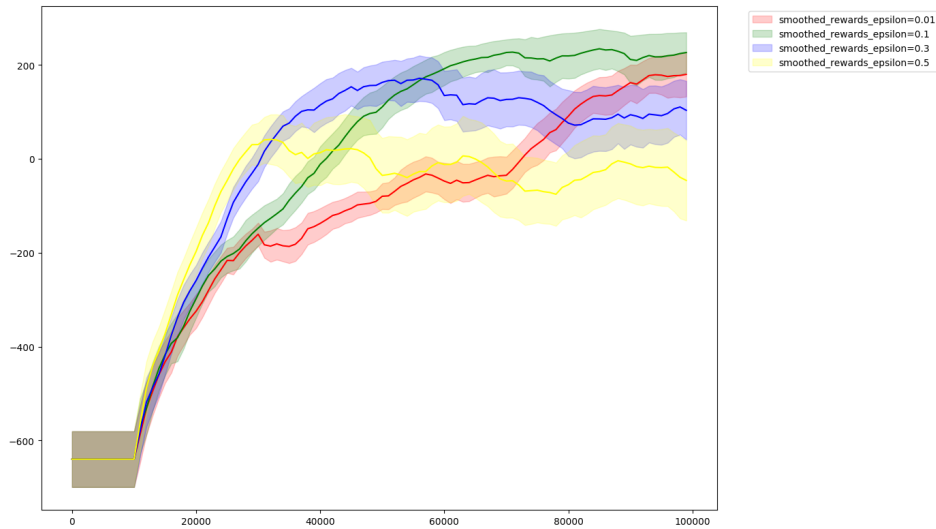


Figure 1: Average episode reward during training of **Epsilon-greedy DQN**.

In this strategy higher ϵ values tend to perform better during initial training, but later they are being outperformed by models trained with lower ϵ values. I believe, this comes out of the fact that at the start data exploration is more beneficial. However later random guesses are becoming more of an obstacle for the model than an opportunity to actually gain more knowledge. Tuning learned patterns becomes then much more useful, and that is why models more focused on data exploitation despite slower starts, later can achieve superior performance.

1.2 Epsilon-greedy with annealing DQN

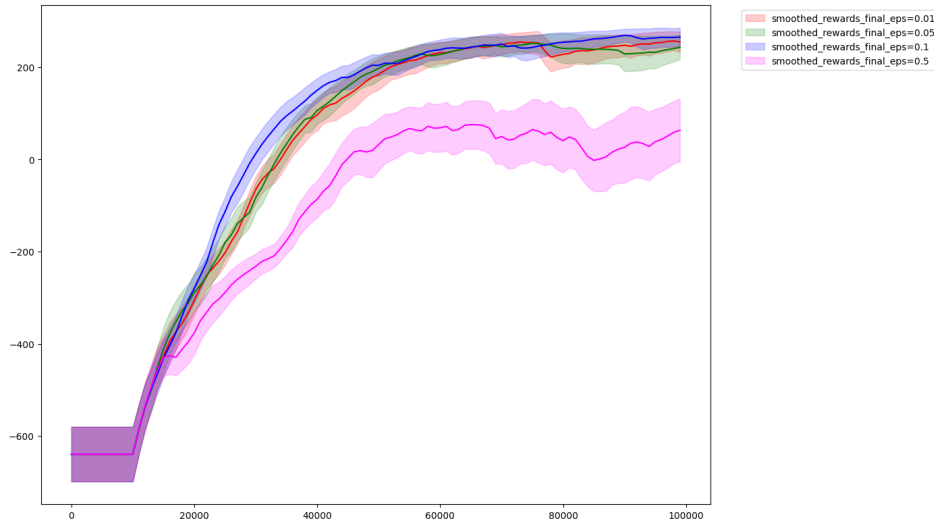


Figure 2: Average episode reward during training of **Epsilon-greedy with annealing DQN**.

I think this confirms my previous conclusions, model stuck with bigger ϵ struggles later to get more insight about optimal paths and is outperformed by models using lower ϵ . The most optimal hyperparameter value here is I believe $\epsilon = 0.1$. It is the most stable during entire training process.

1.3 Boltzmann DQN

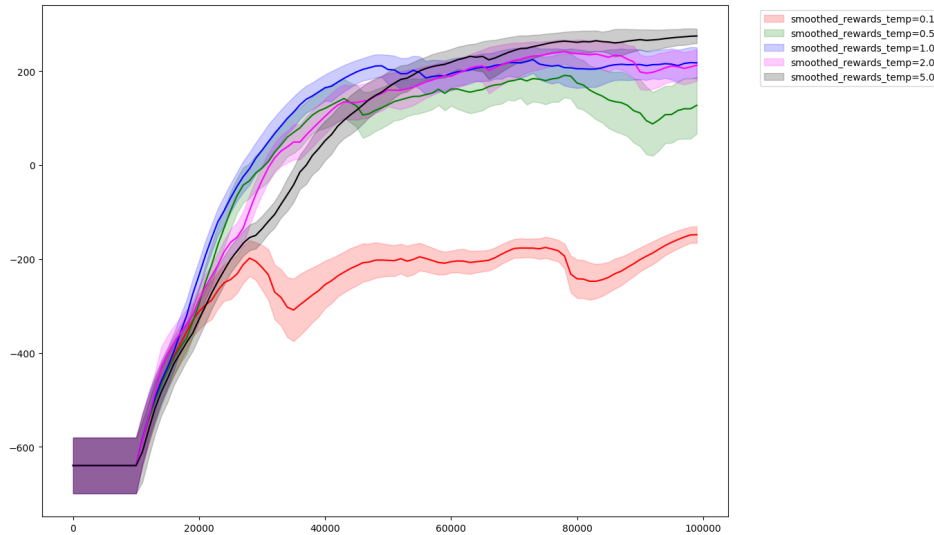


Figure 3: Average episode reward during training of **Boltzmann DQN**.

We can observe that, similarly to previous approaches, models behaving more randomly, i. e. those with lower *temperature*, struggle later since they in lower degree exploit knowledge learned during training. I think higher *temperature* values are more reasonable in this strategy. The highest value, 5, achieves in the end the best scores.

1.4 Max-Boltzmann DQN

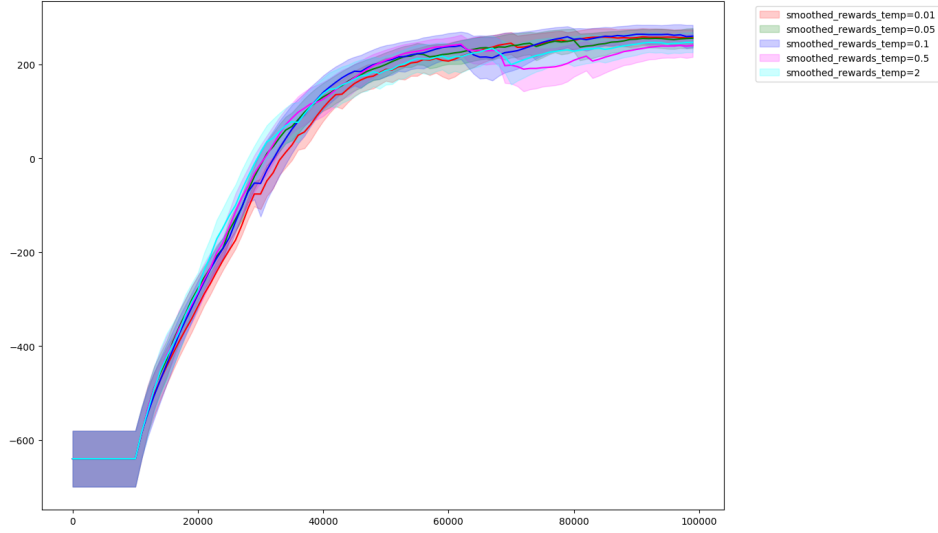


Figure 4: Average episode reward during training of **Max-Boltzmann DQN**, default value $\epsilon = 0.1$.

In this strategy on the other hand changing the *temperature* doesn't seem to be influencing training results in a meaningful way. This is due the fixed $\epsilon = 0.1$, most of the action choices ($\sim 90\%$) are greedy. **Boltzmann** choices are too rare to observe a meaningful influence of *temperature*.

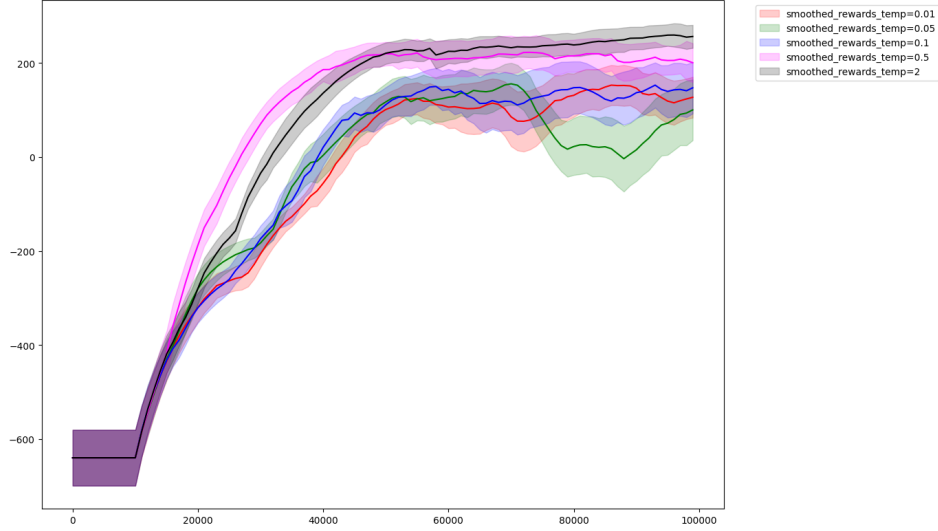


Figure 5: Average episode reward during training of **Max-Boltzmann DQN**, $\epsilon = 0.5$.

Those observations suggest that higher *temperature* for **Max-Boltzmann** strategy make training more stable. The most optimal tested values are 0.5 and 2.

1.5 Max-Boltzmann with temperature annealing DQN

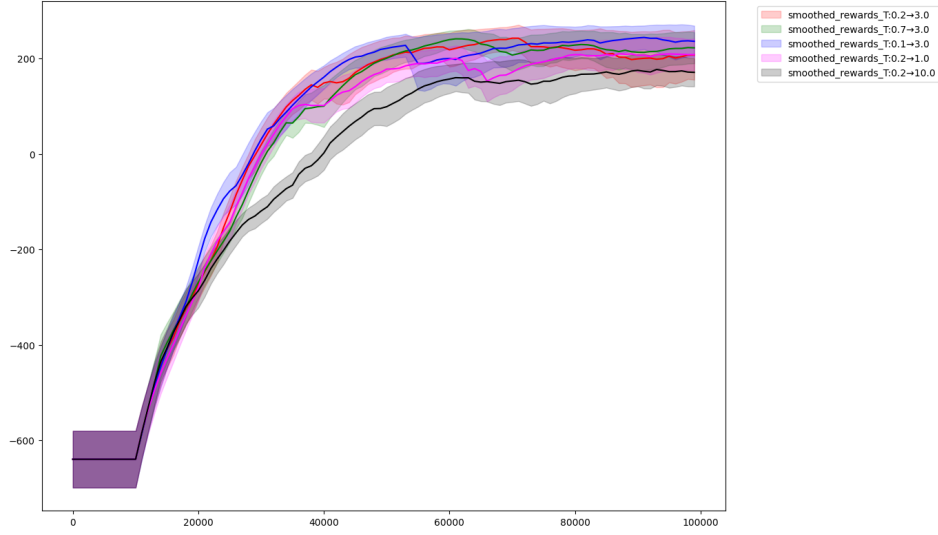


Figure 6: Average episode reward during training of **Max-Boltzmann with temperature annealing DQN**, $\epsilon = 0.5$.

In this strategy I have already changed value of ϵ to 0.5 because, similarly as above, with default ϵ changes after applying different *temperatures* was difficult to observe. Here on the other hand model with the highest *final temperature* seems to be under performing in the final part of training. Probably it didn't had enough time to gain enough knowledge, and too high temperature trapped the model on suboptimal data. The best performance was achieved with the lowest initial *temperature* and average *final temperature*.

1.6 Boltzmann with temperature annealing DQN

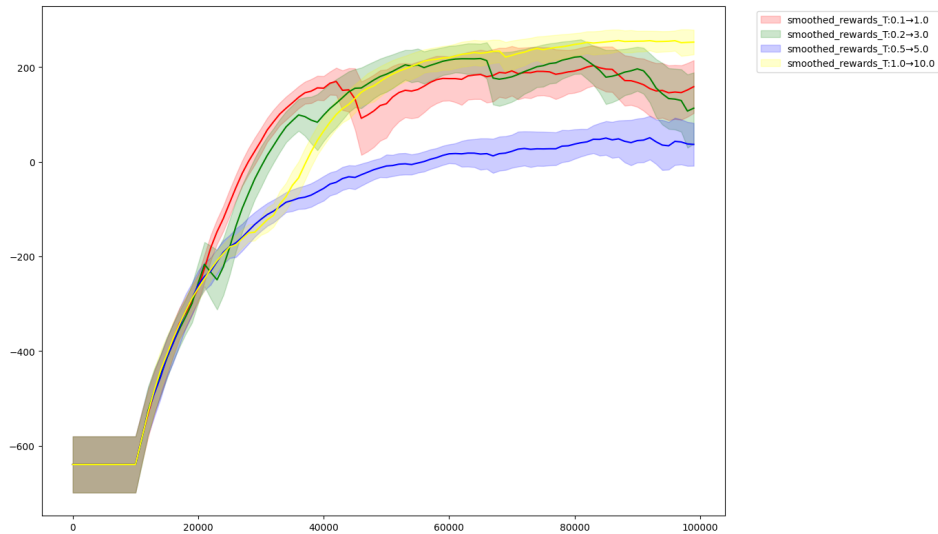


Figure 7: Average episode reward during training of **Boltzmann with temperature annealing DQN**.

In this strategy again higher temperature, meaning more exploitation-focused training, achieves more stable results in the later part of training. However lesser *starting temperature* and *final temperature*

and achieve not much worse results in much quicker time. I believe that the most optimal setting in this strategy might be *starting temperature* = 0.2, *final temperature* = 3 or perhaps 10.

1.7 Combination of epsilon-greedy with epsilon annealing and Boltzmann strategy

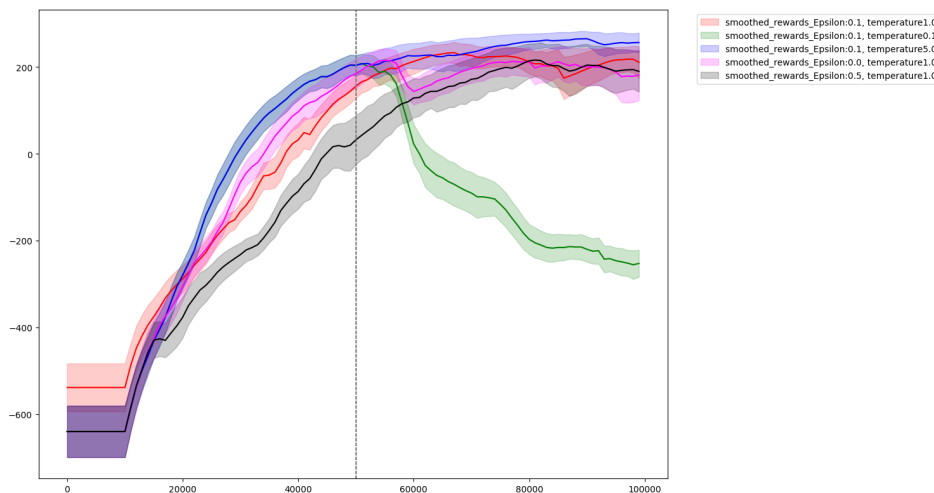


Figure 8: Average episode reward during training of **combination of epsilon-greedy with epsilon annealing and Boltzmann strategy**, with vertical line separating training strategies.

Too low *temperature* completely destroyed good performance of initial training. $\epsilon = 0.1$ in combination with *temperature* = 5 trained really well and high *temperature* allowed the model to well exploit learned state-action pairs.

2 Methods comparison

2.1 Combining two strategies into one

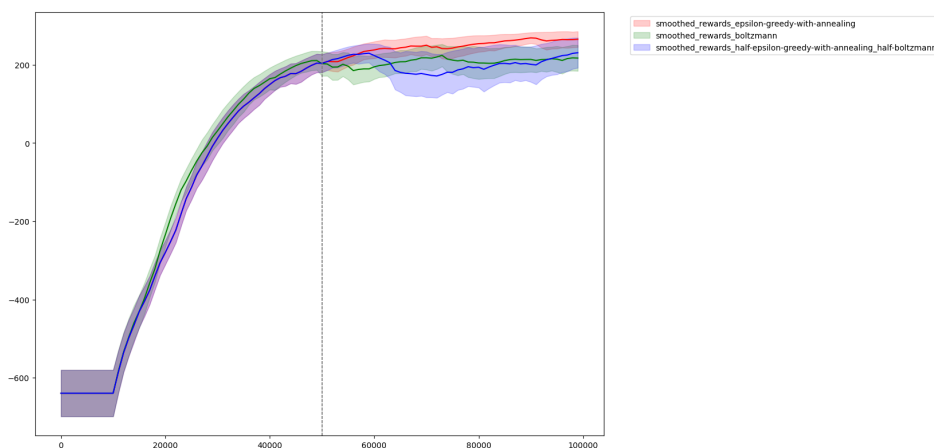
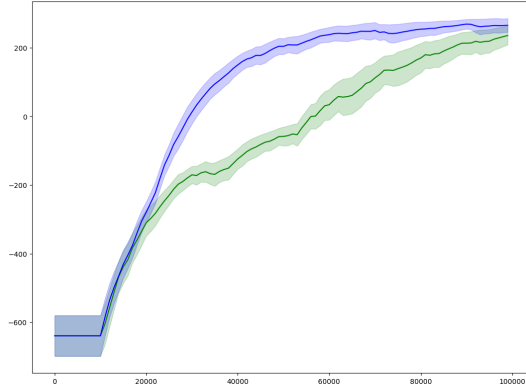


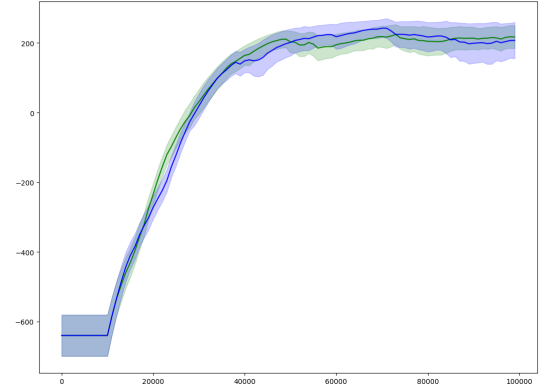
Figure 9: Average episode reward during training of **Epsilon-greedy DQN, Boltzmann DQN and their combination**.

Two methods seem to combine really well. **Combination** curve follows almost exactly curve of the strategy that was used in each half of the training.

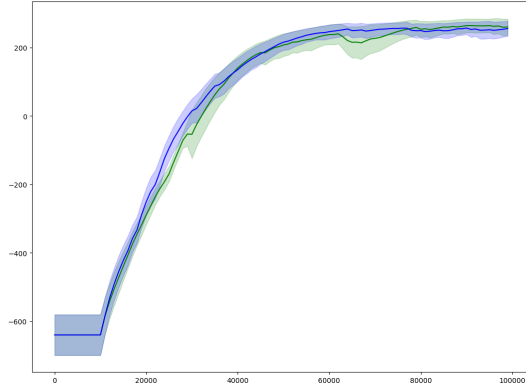
2.2 Annealing vs non-annealing comparison



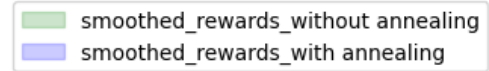
(a) Epsilon-greedy DQN ($\epsilon = 0.1$) vs. Epsilon-greedy with epsilon annealing ($1 \rightarrow 0.1$).



(b) Boltzmann DQN ($t = 1$) vs. Boltzmann with temperature annealing ($0.25 \rightarrow 3$)



(c) Max-Boltzmann DQN ($t = 0.1$) vs. Max-Boltzmann with temperature annealing ($0.025 \rightarrow 0.3$).



(d) Legend

Figure 10: Average episode reward during training for different exploration strategies and their annealed variants. Provided hyperparameters were used.

Annealing the ϵ behaviour of the learning curve much more than annealing *temperature*. However this may be the result of scale of change and value of ϵ used in the models (above we have already determined that temperature change can influence the learning curve in more significant way).

2.3 All strategies

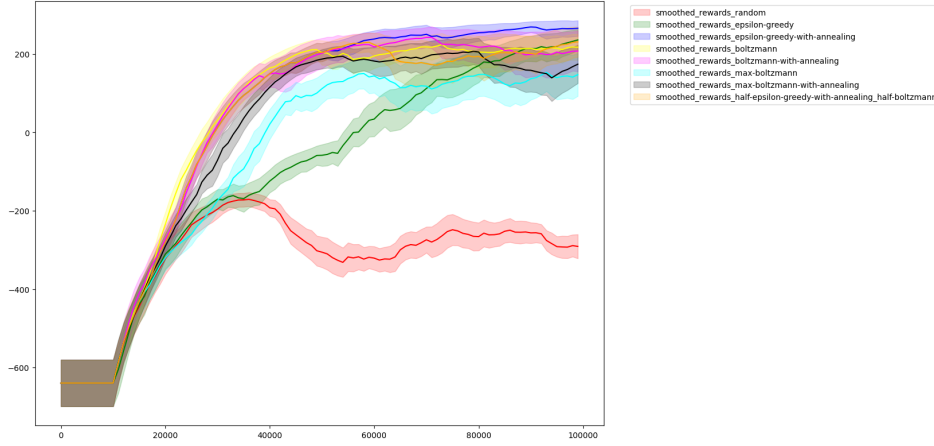


Figure 11: Average episode reward during training of every considered strategy.

Every strategy performed better than the baseline (**random DQN**). The greatest improvement came from introducing ϵ annealing.

3 Bonus - Boltzmann DQN with nonlinear temperature annealing

I implemented two nonlinear ways to anneal temperature: polynomial and logarithmic. I have tested following polynomial degrees: $\frac{1}{2}$, 1 (linear as a reference), and 2. Logarithmic annealing was computed with e as base.

The progress functions were computed using following formulas:

$$\text{progress} = \frac{\text{current step}}{\text{total temperature anneal steps}}$$

$$\text{polynomial progress} = \text{progress}^{\text{polynomial degree}}$$

$$\text{logarithmic progress} = \ln(1 + \text{progress} \cdot (e - 1))$$

Then for the progress function f :

$$\text{temperature difference} = \text{final temperature} - \text{starting temperature}$$

$$\text{current temperature} = \text{starting temperature} + \text{temperature difference} \cdot f(\text{progress})$$

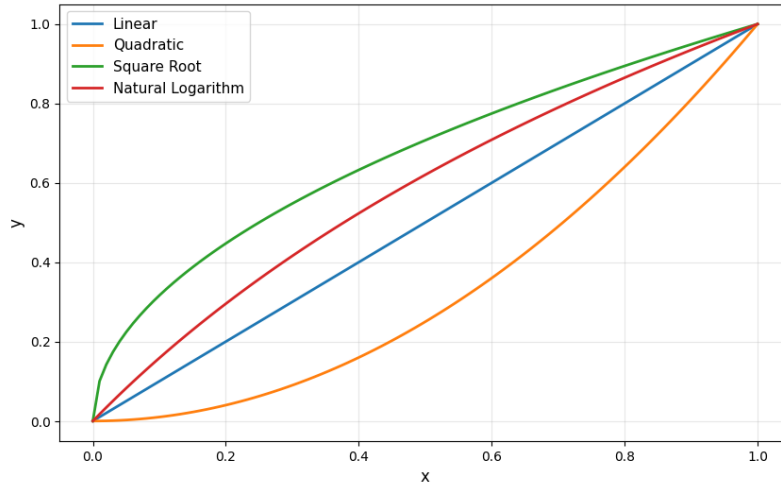


Figure 12: Different progress functions in range $[0, 1]$.

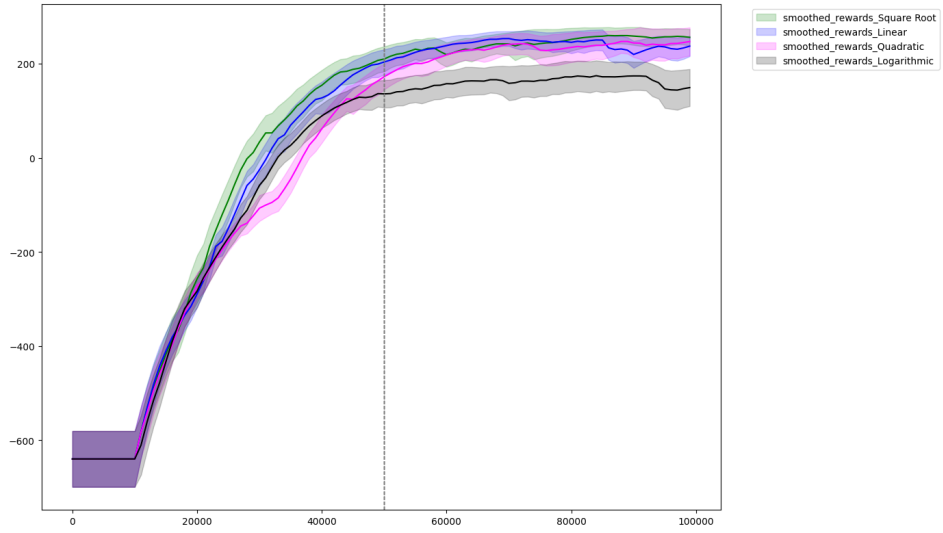


Figure 13: Average episode reward during training of **Boltzmann DQN with non linear temperature annealing**, *temperature anneal steps* was changed to 50000 in order to increase the visibility of differences.

Square root performs the best in the later part of annealing (steps 25000 to 50000) since in this part it increases the temperature in a smoother way than the other functions.